

WEI HONG LOW

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SUMMARY

AI Engineer with 6+ years building and shipping production ML systems, LLM agents, and RAG pipelines. At Binance, built an LLM agent for P2P fraud review that reasons over live transaction and risk data and returns auditable decisions analysts rely on every shift (~90% precision in production). At OKX, built an LLM-as-judge pipeline grading machine translation across 13 language pairs. Strong on agent harness design and rigorous evaluation, with software engineering fundamentals across SQL, streaming (Kafka), and production deployment. M.Sc. in Computer Science (ML), Georgia Tech.

SKILLS

Agentic AI / LLM Systems	: Agents, agent harness design (tool definitions, context assembly, guardrails), multi-skill agent workflows, knowledge-grounded agents, RAG, prompt engineering, LLM evaluation / automated graders, function calling / tool use, LLM-as-a-judge, context retrieval, reasoning workflows
Multimodal & Foundation Models	: Vision-Language Models (VLMs), ALBEF, representation learning, semantic segmentation, multimodal fusion, embeddings, post-training (RLHF, DPO)
LLM / NLP	: Multilingual NLP, machine translation, text classification, transformers (OPUS, NLLB, BERT), information retrieval, OCR
Frameworks / Tools	: PyTorch, Hugging Face, FAISS, OpenSearch, CTranslate2, FastAPI, Pydantic, React / Next.js, LangChain, LlamaIndex, vLLM, SGLang, Anthropic Claude API, Azure OpenAI
Production / Data	: Python, async Python, SQL, PostgreSQL, Redis, Docker, Apache Spark, Hadoop, Apache Airflow, MLflow, AWS SageMaker / S3, Azure AI, Kafka, Git, CI/CD, monitoring / logging
ML Foundations	: Deep learning, reinforcement learning (RLHF, DAgger), optimization, statistical analysis, A/B testing, GNNs

WORK EXPERIENCE

Data Scientist · Community Dream Foundation · Remote	Mar 2026 – present
- Engineered an agentic data collection pipeline for the DreamlineAI initiative to automate daily tracking of government clean energy subsidies. - Integrated Claude as an autonomous agent to conduct targeted web searches, discover new subsidy links, and parse unstructured information into structured, queryable data.	
Senior Data Science Engineer · Binance · Remote	Jan 2026 – Mar 2026
- Built and shipped a production LLM agent for P2P scam review on AWS SageMaker (Flask service + Kafka consumer), processing 600+ dispute reports daily and replacing a binary XGBoost classifier capped at 70% precision. - Engineered the agent’s context-assembly layer: Airflow DAGs pre-aggregate user evidence, N-day transaction history, XGBoost risk scores, and compliance policy into a structured prompt context per report, injected into the Claude API at inference time. - Applied GEPA prompt optimization to produce plain-text reasoning outputs (risk factors, counter-evidence, confidence, recommended action), enabling risk analysts to audit LLM decisions directly rather than acting on opaque classifier scores. - Achieved ~90% precision consistently in production; explainability of LLM output reduced analyst decision time and improved case throughput.	
EIC Lab Research Intern · Georgia Institute of Technology · Atlanta, GA	Mar 2025 – May 2025
- Investigated automated fault localization for Verilog codebases, inspired by SWE-bench, evaluating retrieval strategies (BM25, embedding-based) for surfacing code chunks relevant to GitHub-reported hardware issues. - Built a repo-specific BM25 retrieval pipeline and benchmarked it against alternative methods to identify fault-relevant Verilog snippets, feeding extracted chunks to an LLM for fault hypothesis generation.	
Senior Data Scientist · OKX · Singapore	Nov 2022 – Aug 2024
- Fine-tuned OPUS and NLLB transformer models across 13 language pairs for Web3-domain machine translation, shifting localization teams from full vendor translation to review-only; reduced external translation costs by 35% and cut publication turnaround from days to hours. - Built an automated grader (LLM-as-judge on Azure OpenAI) scoring translations across 4 axes (accuracy, terminology, fluency, locale conventions), enabling systematic model selection and regression checks across all 13 language pairs without manual review. - Deployed a hybrid RAG system for P2P merchant support using FAISS: direct retrieval for high-confidence matches (>80% similarity threshold), falling back to top-5 chunk extraction + LLM reasoning for ambiguous queries; reduced CS team workload by 35%.	

- Built two KYC risk components: (1) OpenSearch vector-based address matching to detect multi-account promotion abuse (95% accuracy); (2) EasyOCR passport validation to reject same-side ID submissions before routing to external verification, reducing unnecessary external API calls (99% accuracy).
- Pioneered a centralized distributed web crawling platform (Crawlab) with proxy management and Redis retry mechanisms, serving 10+ internal teams; built downstream scrapers for competitor intelligence, [X.com](#) complaint monitoring (100% recall on official account replies), and a text classification model routing industry news to correct internal teams (BD, product) at 95%+ accuracy.

Senior NLP Data Scientist · Shopee · Singapore

Jun 2019 – Jun 2022

- Developed and scaled an ALBEF-based multimodal classification model for Indonesian marketplace listings, achieving 80%+ accuracy across 3,000+ complex product categories.
- Spearheaded end-to-end ML lifecycle for text spam detection: engineered automated data pipelines (Airflow, Spark, Hadoop) to extract training data from millions of listings; deployed classification models across 6 international markets with 98%+ precision.
- Engineered text-mining extraction pipelines to identify product keywords and SPU classifications (e.g., laptops), maintaining 97% regional accuracy and securing a 40%+ acceptance rate from internal reviewers.
- Deployed and maintained 7 daily ML pipelines using Airflow for production monitoring and updates.

Business Intelligence Analyst · Shopee · Singapore

Aug 2018 – May 2019

- Single-handedly led the Lowest Price Guarantee (LPG) initiative end-to-end — architecting scalable web crawlers, data pipelines, and databases — improving system efficiency and reducing LPG claim rates by 50%.
- Led and mentored a cross-functional technical team, establishing automated Google Workspace reporting pipelines and converting high-performing interns to full-time engineering roles.

SELECTED PROJECTS

Cosmos Data Gap Analyzer | Agentic training data curation pipeline | github.com/lowweihong/cosmos-data-analyzer

- Designed an agentic pipeline that turns a plain-English failure hypothesis into a validated, targeted dataset for NVIDIA Cosmos — automatically deriving evaluation criteria, generating diverse scenarios, and curating frames via a VLM judge (Claude Sonnet).
- Solved the diversity-vs-relevance tension at two layers: cosine similarity filter at planning (rejects near-identical scenarios) + DSPy metric-driven guardrail at retry (checks rewritten prompts stay on-hypothesis before allowing retry).
- Stack: FLUX.1-dev via NVIDIA NIM, Claude Sonnet 4.6 (VLM judge), DSPy (prompt optimization), SQLite + FTS5 (persistent memory), W&B (observability).

GraphRAG Evaluation Project | [M.Sc.](#) course project, Georgia Tech | github.com/KeXin95/HuggingfaceKG-retriever

- Designed BM25-augmented node features for multi-label classification on a 107K-node Hugging Face Knowledge Graph; lifted PR-AUC by 9.8% (0.360 → 0.395) over baselines.
- Implemented and trained 5 GNN variants over 5 random seeds. Graph Transformer + Focal Loss hit 0.717 Micro-F1, more than doubling tail-class F1 (0.074 → 0.164) over GATv2.

Modified DAgger for CarRacing-v3 | [M.Sc.](#) course project, Georgia Tech | <https://github.com/lowweihong/grp6-dagger>

- Reproduced DAgger on CarRacing-v3 to 98.4% of PPO expert reward (625 vs 635); behavioral cloning baseline collapsed to 8% under distribution shift.
- Extended DAgger with KDE-based trajectory filtering and Gaussian action noise injection — cut training dataset size by 50%+ while matching vanilla DAgger's final reward.
- Library: PyTorch, Stable-Baselines3, Gymnasium

RESEARCH & PUBLICATIONS

H. Huang, W. Low, D. Zhang, et al. (2025). "EIDSeg: A Pixel-Level Semantic Segmentation Dataset for Post-Earthquake Damage Assessment from Social Media Images." Accepted to AAAI 2026 (Artificial Intelligence for Social Impact Track).

EDUCATION

M.Sc. in Computer Science (Machine Learning) · Georgia Institute of Technology · Atlanta, GA

Jan 2025 – Dec 2025

B.Sc. (Hons) in Mathematics and Economics (Double Major) · Nanyang Technological University · Singapore

Jul 2018

CERTIFICATIONS

Aug 2018 NLP with Deep Learning — Stanford Online, Dec 2020

Executive Program in Algorithmic Trading — QuantInsti, Aug 2018

LANGUAGES

Mandarin (Native) · English (Native) · Malay (Intermediate)